

Contents

1	Abstract	3
2	Acknowledgement	3
3	Introduction	3
3.1	Background	3
3.2	Mission and Vision	3
3.3	Objectives and Aims of the Training	3
4	Computed Tomography (CT) System	4
4.1	Theory of CT Scanning	4
4.2	Overall CT System Architecture	4
4.3	CT Gantry and X-ray Tube Unit	4
4.4	Detector and Data Acquisition System (DAS)	4
4.5	Power Distribution Unit (PDU)	5
5	Cyclotron and PET Radiopharmacy	5
5.1	Theory of Medical Cyclotron	5
5.2	Main Cyclotron Subsystems	6
5.3	PET Tracer Synthesis Hotcell and TRACERlab / FASTlab	7
5.4	Helium / Air Compressor and Cooling Water Panel	8
5.5	RF / Power Rack and Compressor Unit	9
6	MRI (1.5T) and Subsystems	9
6.1	Working Principle of MRI	9
6.2	Main MRI Subsystems	10
6.3	MRI Gantry and Console	10
6.4	Cryogenics: Helium Compressor and Cooling System	11
6.5	RF Coils, Gradient Coils, and Imaging System	11
6.6	Preventive Maintenance (PM) for MRI	11
7	Anesthesia Delivery System in OT	12
7.1	Overview of the Anesthesia Workstation	12
7.2	Major Subsystems	12
7.3	Subsystem Illustrations	12
8	Ultrasound System (GE Voluson E10)	13
8.1	Probe Types and Applications	13
8.2	Ultrasound Probe Images	14
8.3	System Features of the GE Voluson E10	14
8.4	Ultrasound Preventive Maintenance (PM)	15
9	Exercise Tolerance Test (ETT) System	15
9.1	GE T2100-ST1 Medical Treadmill	16
9.2	Workflow of the ETT Procedure	17
9.3	Safety Considerations	17
10	Conclusion	17

List of Figures

1	Open view of GE CT gantry showing the rotating frame, tube housing, high-voltage connectors, and cooling components.	5
2	GE CT X-ray Tube Unit, including rotating anode, collimator assembly, and HV cable connections.	5
3	Internal wiring of a GE CT Power Distribution Unit (PDU), showing relays, breakers, fuses, and power routing modules.	6
4	Radiochemistry hotcell showing synthesis hardware, vials, and fluid pathways used for automated PET tracer production under shielded and sterile conditions.	7
5	GE TRACERlab and FASTlab automated synthesis modules for preparation of ^{18}F -labelled radiopharmaceuticals.	7
6	Cooling and process utilities panel with flow meters, valves, and pressure gauges for distribution of chilled water and process gases.	8
7	Target gas and HPLC plumbing panel showing valves, metering pumps, and sample loops for fluorine production and radiochemical purification. .	8
8	RF and power electronics rack with associated gas cylinders and safety interlocks, providing RF power and high-voltage control for beam acceleration.	9
9	Compressor unit providing compressed air / helium for pneumatic actuators, target cooling, and gas transfers in the cyclotron system.	9
10	MRI operator console with GE SIGNA workstation and monitoring panel. The gantry is visible through the RF-shielded glass window.	10
11	MRI power supply and electronic racks containing RF amplifiers, gradient drivers, and system controllers.	10
12	CSA-71 Helium compressor unit (Sumitomo) used for cooling the superconducting magnet and maintaining cryostat pressure.	11
13	MRI RF penetration panel enabling filtered power, control, and RF cabling into the shielded magnet room.	11
14	15" touchscreen CARESCAPE™ monitor B650 displaying ECG, SpO_2 , NIBP and anesthetic gas parameters.	13
15	Modular respiratory monitoring unit supporting low sample flow and integrated gas analysis, designed for easy service.	13
16	Gas chamber and breathing system with reservoir bag, ACGO outlet and safety valves for manual or mechanical ventilation.	13
17	Tec* 7 vaporizers: agent-specific, temperature-compensated units delivering precise concentrations of volatile anesthetic agents.	13
18	Durable work surface with bi-level illumination and flow-control lighting to support safe, efficient clinician workflow.	13
19	Easy-change CO_2 absorbent canister (soda lime / calcium hydroxide) for removal of exhaled CO_2 in circle breathing circuits.	13
20	TVS 4D Probe used for gynecological and early pregnancy imaging. . . .	14
21	GE Voluson E10 ultrasound console with touchscreen and control dial. .	14
22	Probe connector panel supporting multiple transducer ports.	15
23	Linear array probe used for vascular and small-organ imaging.	15
24	GE T2100-ST1 treadmill used for graded exercise testing.	16

1 Abstract

This report summarises the industrial attachment undertaken at GE Healthcare Bangladesh. During the training period, we observed and learned about several advanced medical devices and subsystems, including computed tomography (CT) scanners, cyclotrons, magnetic resonance imaging (MRI) systems, preventive maintenance practices, CT safety procedures, ultrasound machines, exercise tolerance test (ETT) equipment and anaesthesia machines in operating theatres. The report also highlights the preventive maintenance of ultrasonography equipment. The whole attachment creates a new dimension of knowledge where healthcare meets engineering. Throughout the sessions, we underscore GE Healthcare's role in providing cutting-edge solutions for Bangladesh healthcare system.

2 Acknowledgement

We express our heartfelt gratitude to manager, engineers at GE Healthcare Bangladesh for their guidance and support during our industrial attachment. Special thanks are due to our university departmental head for organising the attachment and providing the necessary resources. We appreciate the patience and dedication of the technical staff who explained the complexities of each subsystem and facilitated hands-on learning. Finally, we acknowledge our fellow trainees for their collaboration throughout the training period.

3 Introduction

3.1 Background

The industrial attachment programme at GE Healthcare Bangladesh was designed to bridge the gap between theoretical knowledge and practical application in biomedical engineering. GE Healthcare is a global leader in medical technology, focusing on imaging, diagnostics and patient monitoring systems. The Bangladesh branch serves local hospitals and clinics, diagnostic centers providing installation, maintenance and training for various medical devices.

3.2 Mission and Vision

GE Healthcare's mission is to improve lives by providing innovative medical technologies and services. Its vision emphasises enabling earlier diagnosis, better patient outcomes and more efficient healthcare delivery. During our attachment, we witnessed GE's commitment to reliability, safety and continuous innovation.

3.3 Objectives and Aims of the Training

The objectives of the attachment were:

- To familiarise trainees with the operation, subsystems and maintenance of medical imaging and diagnostic equipment.
- To understand preventive maintenance schedules and safety procedures for critical devices.

- To observe the workflow of healthcare technology management in a professional setting.

4 Computed Tomography (CT) System

4.1 Theory of CT Scanning

Computed Tomography (CT) is an advanced imaging modality that uses a rotating X-ray source and a curved detector array to acquire cross-sectional images of the human body. Unlike conventional radiography, which produces a 2D projection, CT reconstructs volumetric data using algorithms such as Filtered Back Projection (FBP) or iterative reconstruction. [1]

4.2 Overall CT System Architecture

A modern CT system consists of the following major subsystems:

- **Operator Console** – Interface for protocol selection and scan planning.
- **Gantry** – Houses X-ray tube, detector array, slip-ring, collimators, and cooling circuits.
- **Patient Table** – Motorized platform for accurate patient positioning.
- **Power Distribution Unit (PDU)** – Regulates and distributes electrical power.
- **Host Computer / DAS** – Performs data acquisition, control, and image reconstruction.

4.3 CT Gantry and X-ray Tube Unit

The CT gantry is the mechanical core of the system, incorporating:

- High-speed rotating frame with slip-ring for continuous rotation
- X-ray Tube Unit (anode, cathode, rotor, filament)
- Beam-shaping filters and collimators
- Detector array (scintillators + photodiodes)

4.4 Detector and Data Acquisition System (DAS)

The detector array converts incoming X-ray photons into electrical signals. It consists of:

- **Scintillator crystals** (e.g., GOS) converting X-ray energy into light
- **Photodiodes** converting light into current

These signals are transferred to the image reconstruction engine for generating diagnostic images.



Fig. 1: Open view of GE CT gantry showing the rotating frame, tube housing, high-voltage connectors, and cooling components.



Fig. 2: GE CT X-ray Tube Unit, including rotating anode, collimator assembly, and HV cable connections.

4.5 Power Distribution Unit (PDU)

The PDU supplies stable power to all CT subsystems. It contains:

- Circuit breakers
- EMI filters
- Voltage regulators and relay circuits
- Isolation and safety switches

5 Cyclotron and PET Radiopharmacy

5.1 Theory of Medical Cyclotron

A medical cyclotron is a compact particle accelerator used to produce short-lived positron-emitting radioisotopes, most commonly ^{18}F for PET imaging. Charged particles (typically H^+ ions or protons) are injected into a strong magnetic field and accelerated in a spiral path by an alternating radiofrequency (RF) electric field between two “D”-shaped electrodes (dees). When the desired energy is reached, the beam is extracted and directed onto a target containing enriched gas or liquid (e.g. ^{18}O -enriched water), where nuclear reactions produce the required radionuclide. [2]

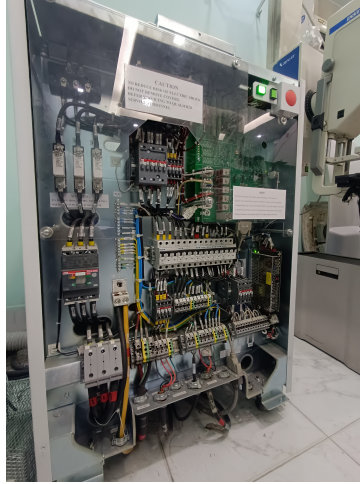


Fig. 3: Internal wiring of a GE CT Power Distribution Unit (PDU), showing relays, breakers, fuses, and power routing modules.

The produced ^{18}F is then transferred to an automated chemistry module to synthesize PET tracers such as ^{18}F -FDG, followed by purification, quality control, and dispensing under aseptic conditions.

5.2 Main Cyclotron Subsystems

Key subsystems of a modern GE medical cyclotron installation include:

- **Magnet and RF system** – Provides magnetic confinement and RF acceleration.
- **Vacuum system** – Maintains low pressure in the cyclotron chamber using roughing and high-vacuum pumps.
- **Gas handling and targetry** – Manages ^{18}O -water, fluorine gas or other target materials.
- **Helium / air compressor system** – Supplies compressed gas for pneumatic valves, target cooling, and transfers.
- **Cooling water system** – Removes heat from magnet, RF amplifiers, targets, and power electronics.
- **PET tracer synthesis modules** – Automated modules for radiolabelling (TRACERlab, FASTlab).
- **Purification and QC** – HPLC, sterile filtration, and analytical instruments.

5.3 PET Tracer Synthesis Hotcell and TRACERlab / FASTlab

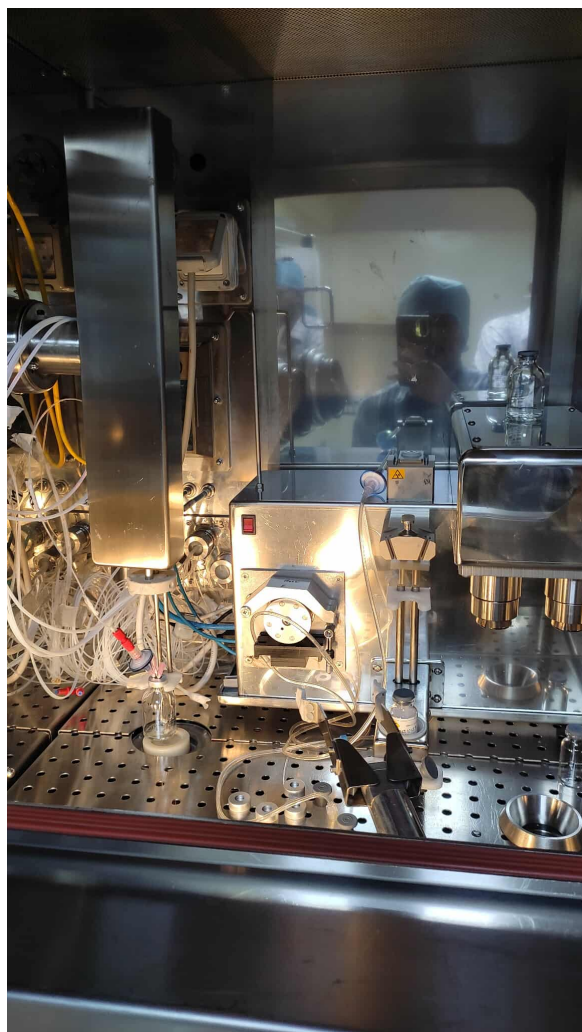


Fig. 4: Radiochemistry hotcell showing synthesis hardware, vials, and fluid pathways used for automated PET tracer production under shielded and sterile conditions.



Fig. 5: GE TRACERlab and FASTlab automated synthesis modules for preparation of ^{18}F -labelled radiopharmaceuticals.

The synthesis hotcell provides a shielded enclosure for handling high-activity solutions. GE TRACERlab and FASTlab units automate multiple steps: drying the ^{18}F fluoride, nucleophilic substitution, hydrolysis, neutralization, and sterile formulation, thereby reducing operator dose and improving reproducibility.

5.4 Helium / Air Compressor and Cooling Water Panel



Fig. 6: Cooling and process utilities panel with flow meters, valves, and pressure gauges for distribution of chilled water and process gases.

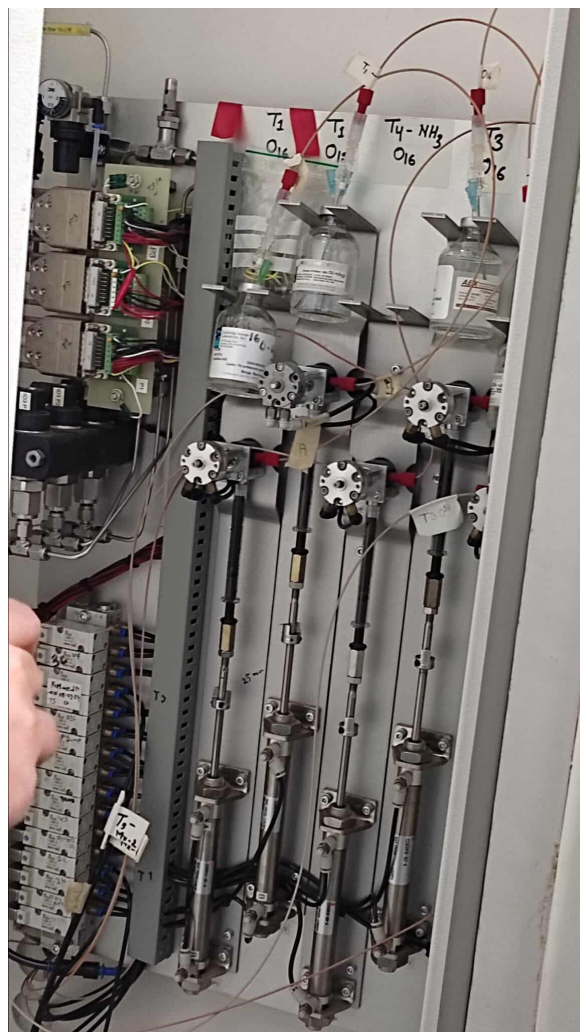


Fig. 7: Target gas and HPLC plumbing panel showing valves, metering pumps, and sample loops for fluorine production and radiochemical purification.

The cooling water panel distributes temperature-controlled water to the magnet coils, RF amplifiers, vacuum pumps, and target assemblies, ensuring thermal stability and reliability. The target / HPLC panel interfaces between the cyclotron targetry and the radiochemistry module. After irradiation, the radioactive solution is transferred through shielded lines into the synthesis cassette and, where required, through an HPLC column for purification, followed by sterile filtration before patient use.

5.5 RF / Power Rack and Compressor Unit



Fig. 8: RF and power electronics rack with associated gas cylinders and safety interlocks, providing RF power and high-voltage control for beam acceleration.



Fig. 9: Compressor unit providing compressed air / helium for pneumatic actuators, target cooling, and gas transfers in the cyclotron system.

The RF / power rack houses amplifiers, RF generators, and control modules responsible for maintaining the correct accelerating voltage and frequency in the cyclotron dees. Adjacent gas cylinders and regulators feed target gases and cooling lines. The compressor maintains the pressure required for pneumatic valves and target cooling jets.

6 MRI (1.5T) and Subsystems

Magnetic Resonance Imaging (MRI) is a non-ionizing imaging technique that uses a strong static magnetic field, gradient fields, and radiofrequency (RF) pulses to generate high-resolution anatomical images. During our industrial training at GE Healthcare Bangladesh, we observed a complete 1.5 Tesla MRI system along with its associated support infrastructure. [2]

6.1 Working Principle of MRI

MRI relies on the alignment of hydrogen protons in a strong magnetic field. When an RF pulse is applied at the Larmor frequency, these protons absorb energy and later release it as they relax back to equilibrium. The detected signals are transformed from *k-space* to image space using the Fourier transform.

Key MRI parameters:

- **TR** – Repetition Time
- **TE** – Echo Time
- **Flip Angle** – Determines excitation strength
- **T1/T2 Relaxation** – Tissue-specific relaxation times

6.2 Main MRI Subsystems

A 1.5T MRI scanner consists of several integrated subsystems:

- **Superconducting Magnet** – Maintains a stable 1.5T magnetic field using liquid helium.
- **Gradient Coils** – Provide spatial encoding along X, Y, and Z axes.
- **RF Coils** – Transmit RF pulses and receive the MR signal.
- **Gantry and Bore** – Houses magnet, RF, and gradient hardware.
- **Helium Compressor / Cryocooler** – Re-condenses helium to reduce boil-off.
- **MRI Console and Host Computer** – Controls acquisition and reconstruction.
- **Power and RF Amplifier Racks** – Drive RF and gradient amplifiers.
- **Room Shielding / Penetration Panel** – RF-shielded room with filtered electrical feed-throughs.

6.3 MRI Gantry and Console



Fig. 10: MRI operator console with GE SIGNA workstation and monitoring panel. The gantry is visible through the RF-shielded glass window.

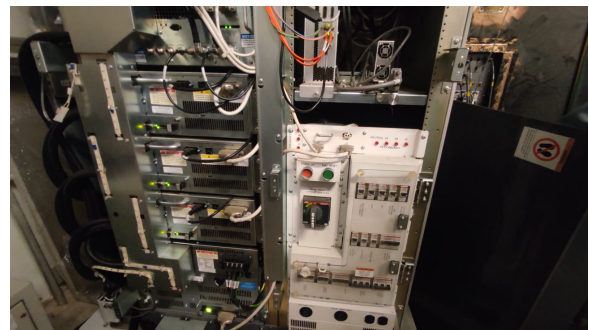


Fig. 11: MRI power supply and electronic racks containing RF amplifiers, gradient drivers, and system controllers.

The MRI gantry houses the superconducting magnet, gradient coils, RF coils, and cryogenic components. The console room provides a safe, shielded location for technologists to operate the MRI scanner and monitor patient safety.

6.4 Cryogenics: Helium Compressor and Cooling System



Fig. 12: CSA-71 Helium compressor unit (Sumitomo) used for cooling the superconducting magnet and maintaining cryostat pressure.

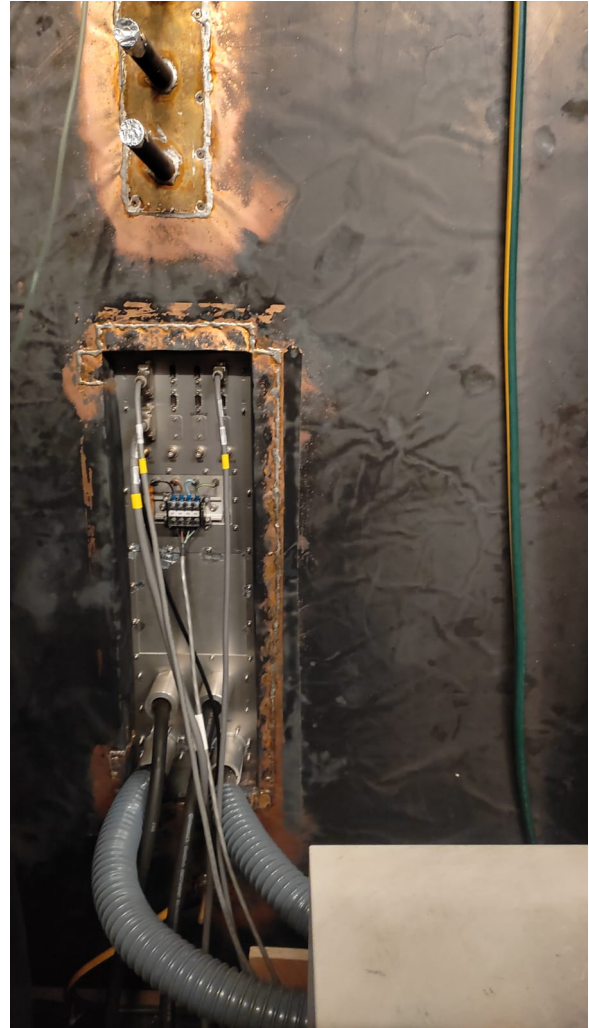


Fig. 13: MRI RF penetration panel enabling filtered power, control, and RF cabling into the shielded magnet room.

The helium compressor circulates high-pressure helium gas to the coldhead (cryocooler), reducing liquid helium evaporation and ensuring stable superconducting operation.

The penetration panel maintains RF shielding integrity while allowing electrical and data lines into the MRI room.

6.5 RF Coils, Gradient Coils, and Imaging System

RF coils act as both transmitters and receivers, tuned to the Larmor frequency of hydrogen. Gradient coils generate small linear variations in magnetic field strength, enabling slice selection and spatial encoding.

6.6 Preventive Maintenance (PM) for MRI

Preventive maintenance activities observed:

- Checking helium level, coldhead performance, and compressor pressures.
- Inspecting gradient amplifiers and RF chain for overheating or noise.
- Verifying bore ventilation, room temperature, and emergency quench lines.
- Cleaning filters in air handling and cooling systems.
- Running image-quality phantoms (SNR, uniformity, ghosting tests).
- Ensuring proper functioning of patient table motors and safety interlocks.

Regular PM prevents downtime, reduces risk of quench, and ensures consistent clinical image quality.

7 Anesthesia Delivery System in OT

7.1 Overview of the Anesthesia Workstation

The GE *Carestation*TM 650 anesthesia delivery system integrates gas delivery, ventilation, patient monitoring and safety features into a single workstation. [3]

7.2 Major Subsystems

Key subsystems of the anesthesia workstation include:

- Patient monitoring and 15" touchscreen display.
- Modular respiratory monitoring and ventilator controls.
- Gas chamber and breathing circuit (ACGO, bags, valves).
- Agent-specific vaporizers (e.g. Isoflurane, Sevoflurane).
- Illuminated, durable work surface and control dials.
- Easy-change CO₂ absorbent canister and storage drawers.

7.3 Subsystem Illustrations



Fig. 14: 15" touchscreen CARESCAPE™ monitor B650 displaying ECG, SpO₂, NIBP and anesthetic gas parameters.



Fig. 15: Modular respiratory monitoring unit supporting low sample flow and integrated gas analysis, designed for easy service.



Fig. 16: Gas chamber and breathing system with reservoir bag, ACGO outlet and safety valves for manual or mechanical ventilation.



Fig. 17: Tec* 7 vaporizers: agent-specific, temperature-compensated units delivering precise concentrations of volatile anesthetic agents.



Fig. 18: Durable work surface with bi-level illumination and flow-control lighting to support safe, efficient clinician workflow.



Fig. 19: Easy-change CO₂ absorbent canister (soda lime / calcium hydroxide) for removal of exhaled CO₂ in circle breathing circuits.

8 Ultrasound System (GE Voluson E10)

Ultrasound imaging uses high-frequency sound waves to generate real-time anatomical images. The GE Voluson E10 is a ultrasound platform specifically optimized for women's health, with advanced support for obstetrics, gynecology, fetal cardiac imaging, and 4D volumetric acquisition. [4]

The E10 supports Matrix 4D probes capable of acquiring four simultaneous imaging planes at very high frame rates.

8.1 Probe Types and Applications

The probes used in the Voluson E10 system include:

- **Convex Probe** – Standard abdominal imaging, obstetrics.
- **Convex 4D Probe** – High-resolution 4D fetal imaging.
- **TVS 4D Probe** – Transvaginal volumetric probe for gynecology and early pregnancy.
- **Linear Probe** – Peripheral vascular, small parts, pediatric applications.

8.2 Ultrasound Probe Images



Fig. 20: TVS 4D Probe used for gynecological and early pregnancy imaging.



Fig. 21: GE Voluson E10 ultrasound console with touchscreen and control dial.

8.3 System Features of the GE Voluson E10

- **Matrix 4D Technology:** Captures four volumes simultaneously at high frame rates, producing realistic 4D fetal images.
- **Advanced Women’s Health Imaging:** Optimized for OB/GYN including fetal heart, fetal growth, uterine assessment, adnexa, and ovarian morphology.
- **High-Performance Processing:** Windows 10 platform with 1 TB SSD storage and rapid data access for large 4D datasets.
- **Display and Interface:**
 - 23-inch primary high-definition display.
 - 12-inch touchscreen control panel for fast parameter adjustment.
- **Comprehensive Probe Support:** Convex, 4D convex, TVS 4D, linear, pediatric, small parts, and specialized OB probes.
- **Clinical Applications:** Obstetrics, gynecology, fetal cardiac imaging, abdominal imaging, thyroid, breast, vascular studies, and pediatric scans.



Fig. 22: Probe connector panel supporting multiple transducer ports.



Fig. 23: Linear array probe used for vascular and small-organ imaging.

8.4 Ultrasound Preventive Maintenance (PM)

Routine PM tasks include:

- Testing transducer integrity and replacing damaged cables.
- Cleaning and disinfecting probes (especially TVS probes).
- Checking system ventilation and fans to prevent overheating.
- Verifying display calibration and touchscreen responsiveness.

9 Exercise Tolerance Test (ETT) System

The Exercise Tolerance Test (ETT), also known as the treadmill stress test, evaluates the functional capacity of the heart under controlled physical exertion. It is commonly used to diagnose coronary artery disease, assess cardiac arrhythmias, and determine exercise-induced symptoms. [5]

A complete GE ETT system includes the following components:

- **ECG lead set** – Multilead ECG cable used for continuous cardiac monitoring.
- **ETT workstation / ECG console** – Displays real-time ECG, stores reports, and controls treadmill operation.
- **Treadmill (GE T2100-ST1)** – Medically rated, computer-controlled treadmill that adjusts speed and incline automatically.

The ECG lead set includes RA, LA, RL, LL, and chest leads used for continuous cardiac rhythm monitoring throughout the stress protocol. High-accuracy filtering minimizes motion artifact during exercise.

The GE CASE or MAC-series ETT workstation provides:

- 12-lead ECG acquisition and real-time display;
- automated ST-segment analysis;
- arrhythmia detection;
- data storage and report generation.

9.1 GE T2100-ST1 Medical Treadmill



Fig. 24: GE T2100-ST1 treadmill used for graded exercise testing.

The GE T2100-ST1 treadmill is designed for medical stress testing. Its features include:

- soft start and smooth speed transition;
- programmable incline and speed control;
- high-torque AC drive system;
- low-profile deck to enhance patient safety;
- operation possible even where 220 V mains power is not available (supports lower-power environments).

9.2 Workflow of the ETT Procedure

1. Patient preparation and ECG lead placement.
2. Baseline ECG, blood pressure, and oxygen saturation recording.
3. Progressive increase of treadmill speed and incline using the selected protocol.
4. Continuous ECG and symptom monitoring.
5. Recovery-phase monitoring until vital signs return to baseline.
6. Automated report generation, including ST-segment changes and heart-rate response.

9.3 Safety Considerations

- Continuous supervision by trained personnel.
- Emergency stop switch and handrails on the treadmill.
- Defibrillator and crash cart available nearby.
- Patient screening for contraindications (recent myocardial infarction, severe hypertension, unstable angina, etc.).

10 Conclusion

The industrial attachment at GE Healthcare Bangladesh provided invaluable and significant hands-on experience with advanced medical imaging equipment and support systems. We gained a comprehensive understanding of CT, cyclotron, MRI, ultrasound, ETT and anaesthesia technologies, along with the importance of preventive maintenance and safety protocols.

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